

OFFICIAL LINKS

 Website

<https://braintube.site>

 WhatsApp Channel (VTU Notes & Updates)

<https://wa.me/918884624741>

 YouTube Channel

<https://youtube.com/@braintube-v9w?si>

BMATE101 – MATHEMATICS-I (EEE)

MODULE-5 : LINEAR ALGEBRA

MATRICES, RANK & SYSTEM OF LINEAR EQUATIONS

1. Matrices

Definition

A **matrix** is a rectangular array of numbers arranged in rows and columns.

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

2. Types of Matrices

- **Row matrix:** one row
- **Column matrix:** one column
- **Square matrix:** rows = columns
- **Zero matrix:** all elements zero
- **Diagonal matrix:** non-zero elements only on main diagonal
- **Identity matrix:**

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

3. Matrix Operations

Addition

Defined only for matrices of same order.

$$A + B = [a_{ij} + b_{ij}]$$

Multiplication

Possible when columns of A = rows of B .

$$(AB)_{ij} = \sum a_{ik} b_{kj}$$

4. Elementary Row Transformations (ERT)

1. $R_i \leftrightarrow R_j$ (Row interchange)
2. $R_i \rightarrow kR_i, k \neq 0$ (Scalar multiplication)
3. $R_i \rightarrow R_i + kR_j$

ERTs are used to find **rank** and solve equations.

5. Rank of a Matrix

Definition

The **rank** of a matrix is the maximum number of **linearly independent rows or columns**.

Method

- Reduce matrix to **row echelon form**
- Count non-zero rows

Example 1

Find rank of:

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

Apply $R_2 \rightarrow R_2 - 2R_1$:

$$\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$$

Rank(A) = 1

6. System of Linear Equations

General form:

$$AX = B$$

where

$A \rightarrow$ coefficient matrix

$X \rightarrow$ variable matrix

$B \rightarrow$ constant matrix

7. Consistency of System

A system is:

- **Consistent** if solution exists
- **Inconsistent** if no solution exists

Rouché–Capelli Theorem

A system is consistent **iff**:

$$\boxed{\text{Rank}(A) = \text{Rank}([A \mid B])}$$

- If rank = number of variables $\rightarrow$ **unique solution**
- If rank < number of variables $\rightarrow$ **infinite solutions**

8. Gauss Elimination Method

Steps

1. Write augmented matrix $[A \mid B]$
2. Apply ERTs to convert into **upper triangular form**
3. Solve by back-substitution

Example 2

Solve:

$$\begin{aligned}x + y &= 2 \\x - y &= 0\end{aligned}$$

Augmented matrix:

$$\left[\begin{array}{cc|c} 1 & 1 & 2 \\ 1 & -1 & 0 \end{array} \right]$$

Apply $R_2 \rightarrow R_2 - R_1$:

$$\left[\begin{array}{cc|c} 1 & 1 & 2 \\ 0 & -2 & -2 \end{array} \right]$$
$$y = 1, x = 1$$

9. Gauss–Jordan Method

Difference from Gauss Elimination

- Reduces matrix to **diagonal form**
- No back-substitution needed

Steps

1. Convert $[A \mid B]$ to diagonal form using ERTs
2. Read solution directly

EIGENVALUES, EIGENVECTORS & RAYLEIGH POWER METHOD

10. Eigenvalues and Eigenvectors

Definition

For a square matrix A , a non-zero vector X is called an **eigenvector** if:

$$AX = \lambda X$$

where λ is a scalar called the **eigenvalue**.

11. Finding Eigenvalues

Rewriting:

$$(A - \lambda I)X = 0$$

For non-trivial solutions:

$$\det(A - \lambda I) = 0$$

This equation is called the **characteristic equation**.

Example 1

Find eigenvalues of:

$$\begin{aligned} A &= \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \\ \det \begin{bmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{bmatrix} &= 0 \\ (2 - \lambda)^2 - 1 &= 0 \\ \lambda^2 - 4\lambda + 3 &= 0 \\ \lambda &= 1, 3 \end{aligned}$$

12. Finding Eigenvectors

For each eigenvalue λ , solve:

$$(A - \lambda I)X = 0$$

Example 2

For $\lambda = 3$:

$$A - 3I = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$$

Solving:

$$X = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

(Any non-zero scalar multiple is also an eigenvector.)

13. Properties of Eigenvalues

- Sum of eigenvalues = trace of matrix
- Product of eigenvalues = determinant of matrix
- Eigenvalues may be real or complex
- Eigenvectors corresponding to distinct eigenvalues are linearly independent

14. Rayleigh Power Method

Purpose

The **Rayleigh power method** is used to find the **dominant (largest) eigenvalue** and its eigenvector.

Procedure

1. Choose an initial vector X_0
2. Compute:

$$X_{k+1} = AX_k$$

3. Normalize X_{k+1}
4. Repeat until convergence
5. Dominant eigenvalue:

$$\lambda \approx \frac{\text{largest component of } AX}{\text{largest component of } X}$$

Example 3 (Conceptual)

Given matrix A , repeated multiplication aligns the vector along the direction of the **largest eigenvalue**.

15. Applications of Eigenvalues

- Electrical network analysis
- Stability analysis
- Vibration problems
- Image processing
- Data science and PCA

16. Self-Study Topics (As per Syllabus)

(Only theory awareness required)

- Gauss–Jacobi Method
- Cayley–Hamilton Theorem